FORM CODE 2F

SECTION 1

Time—30 minutes

40 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. AGITATION: (A) originality (B) curiosity (C) serenity (D) intelligence (E) significance

2. MOBILE: (A) fixed (B) perceptive (C) excluded (D) simplified (E) basic

3. JUNCTURE: (A) separation (B) complexity (C) assumption (D) offensive comment (E) valuable merchandise

4. DETAIN: (A) amuse (B) flatter (C) expedite (D) display clearly (E) invest money

5. STATELY: (A) remote (B) final (C) unusual (D) impractical (E) humble

6. SPURN: (A) react (B) inform (C) embrace (D) bring order to (E) remain firm

7. CONSEQUENTIAL: (A) repellent (B) trivial (C) unauthorized (D) bringing bad luck (E) not easily attained

8. INSULARITY: (A) broadmindedness (B) bluntness (C) courage (D) loss of freedom (E) complexity of design

9. REND: (A) thicken (B) display (C) unify (D) moisten repeatedly (E) contaminate accidentally

10. EXIGENT: (A) lacking urgency (B) without permission (C) inviting interest (D) unfashionable (E) unceremonious

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ---, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

11. Although we often use the terms porpoise and dolphin ----, careful marine biologists observe the ---- the two.

(A) carelessly. .behavior of
(B) consciously. .similarity between
(C) definitively. .habitats of
(D) interchangeably. .distinctions between
(E) accurately. .history of

12. Either the sunsets at Nome are ----, or the one I saw was a poor example.

(A) gorgeous (B) overrated (C) unobserved (D) exemplary (E) unappreciated

13. He had a delightfully indulgent way of showing his ---- for his friends; these actions in themselves ---- a kind heart.

(A) respect. .contradicted
(B) concern. .deprecated
(C) disdain. .established
(D) intolerance. .denoted
(E) fondness. .betokened

GO ON TO THE NEXT PAGE

Copyright © 1982 by Educational Testing Service. All rights reserved.
Princeton, N.J. 08541

14. Ridges on the fingertips are thought to increase ---- in gripping by ---- friction at the surface of contact.

(A) leverage. .converting
(B) efficiency. .creating
(C) strength. .soothing
(D) endurance. .demonstrating
(E) independence. .spreading

15. Fortunately, ---- is one of the bad habits in writing that has gradually gone out of fashion since the daily newspaper first set a standard of ---- reporting of events.

(A) imprecision. .swift
(B) superficiality. .routine
(C) irreverence. .detailed
(D) redundancy. .documented
(E) circumlocution. .concise

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

16. CONCERT : AUDIENCE :: (A) restaurant : waiters
(B) orchestra : musicians (C) game : spectators
(D) school : cheerleaders (E) zoo : keepers

17. WATER : RESERVOIR :: (A) oil : fuel
(B) money : bank (C) lake : ocean
(D) parent : family (E) beach : inlet

18. WALK : PROWL ::
(A) cheat : pretend
(B) speak : shout
(C) applaud : disapprove
(D) listen : eavesdrop
(E) smile : grin

19. STAGNANT : MOTION :: (A) arid : moisture
(B) morbid : dread (C) neutral : balance
(D) marred : fault (E) tilled : irrigation

20. RETOUCH : PICTURE :: (A) design : dress
(B) orchestrate : song (C) publish : magazine
(D) emend : text (E) concoct : pastry

21. CAMOUFLAGE : APPEARANCE ::
(A) compensate : payment
(B) decipher : code
(C) experiment : hypothesis
(D) bluff : intention
(E) invest : chance

22. BIRD : MOLT :: (A) bear : hibernate
(B) snake : slough (C) fish : catch
(D) hawk : prey (E) rabbit : trap

23. INTROSPECTIVE : SELF ::
(A) pompous : thoughts
(B) conceited : others
(C) miserly : accomplishments
(D) impetuous : decisions
(E) scrupulous : principles

24. INCITE : SEDITIONIST :: (A) parade : heckler
(B) assault : victor (C) abdicate : autocrat
(D) arbitrate : mediator (E) donate : financier

25. NOISE : DIN :: (A) injury : pride
(B) voice : speech (C) sincerity : homage
(D) emotion : fervor (E) signal : message

GO ON TO THE NEXT PAGE

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

April 23rd to July 5th, 1832: The climate, during the months of May and June, the beginning of winter, was delightful. The mean temperature, from observations taken at nine o'clock, both morning and evening, was only 72°. It often rained heavily, but the drying southerly winds soon again rendered the walks pleasant. One morning, in the course of six hours, 1.6 inches of rain fell. As this storm passed over the forests which surround the mountain, the sound produced by the drops pattering on the countless multitude of leaves was very remarkable; it could be heard at the distance of a quarter of a mile, and was like the rushing of a great body of water. After the hotter days, it was delicious to sit quietly in the garden and watch the evening pass into night. Nature, in these climes, chooses vocalists from more humble performers than in Europe. A small frog, of the genus *Hyla*, sits on a blade of grass about an inch above the surface of the water, and sends forth a pleasing chirp: when several are together they sing in harmony on different notes. I had some difficulty in catching a specimen of this frog. The genus *Hyla* has its toes terminated by small suckers; and I found this animal could crawl up a pane of glass, when placed absolutely perpendicular. Various cicadae and crickets, at the same time, keep up a ceaseless shrill cry, but which, softened by the distance, is not unpleasant. Every evening after dark this great concert commenced; and often have I sat listening to it, until my attention has been drawn away by some curious passing insect.

26. According to the passage, the sound of the rain was "very remarkable" (lines 11-12) because of its

(A) duration
(B) dissonance and shrillness
(C) loudness and power
(D) fresh, rejuvenating associations
(E) eerie, mysterious quality

27. For a human being to perform a feat comparable to that performed by the frog (lines 25-26), that human being would have to

(A) climb to the top of an icy trail on skis
(B) walk up the side of a building
(C) slide across thin ice without cracking it
(D) climb to the top of a tall shade tree
(E) walk barefoot across a thin wire

28. In the passage, the author exhibits all of the following attitudes toward the surroundings EXCEPT

(A) curiosity
(B) wonder
(C) pleasure
(D) anxiety
(E) objectivity

Ann Beattie has done more than write short stories and two novels; within six short years she has become the voice of the contemporary United States. Of course, in a nation that touts the writing of the Great American Novel as one of its most chimerical achievements, it is preposterous to suppose that one author could mean all things to all people at all times. Instead, it seems that American fiction moves by decades; every ten years there emerges a new writer who is not only perfectly attuned to his or her times, but who can precisely transmit the essence of that era onto the page—not just documentation, but illumination as well. The work of F. Scott Fitzgerald illuminates the 1920's; the 1950's belong to J. D. Salinger.

With great wit and a hawk's eye for detail, Beattie writes about the generation that survived the Vietnam era, the drug and sexual revolutions, and came to rest, burned out, in the seventies. Her people are rootless. They have severed their ties with the old traditions, yet, to their bewilderment, they find themselves dissatisfied with the new "traditions" as well: communal living, unstructured child-rearing, absence of career goals. Their thoughts are as disjointed as their lives. These are not men and women grappling with great moral issues; these are people simply waiting, looking for some measure of satisfaction, some happiness. The problem is, more often than not, that they do not know where to begin to look, and, as is typical of the "me generation," their search frequently entails hurting others.

29. The word "chimerical" as used in line 6 means

(A) elusive (B) contemptible (C) innovative (D) significant (E) original

30. According to information presented in the passage, which of the following best describes a typical Beattie character?

(A) Experimental and relentlessly revolutionary
(B) Creative and refreshingly open-minded
(C) Discontented and often insensitive to others
(D) Goal-oriented, but not dependent on old traditions
(E) Uncertain, but optimistic and unselfish

31. The style and content of this passage indicate that it is most likely an excerpt from

(A) a plot summary on a book jacket
(B) a study of great American novels
(C) a brief biographical sketch of Beattie
(D) a paper on life-styles in the seventies
(E) an article on Beattie and her works

GO ON TO THE NEXT PAGE

The reading passages in this test are brief excerpts or adaptations of excerpts from published material. The ideas contained in them do not necessarily represent the opinions of the College Board or Educational Testing Service. To make the text suitable for testing purposes, we may in some cases have altered the style, contents, or point of view of the original.

1

In the early years of the Third French Republic (1870-1940), an urgent need was felt for an adequate patron, preferably drawn from the great leaders of
Line the First Republic. Danton seemed promising
(5) material and, after a band of official historians had gotten to work on him, his statue glared forth with audacity from its pedestal. Respected historian Albert Mathiez, however, set to work to attack this new idol and, after twenty-four years of research,
(10) made it pretty clear that Danton was unprincipled, venal, implicated in treasonable negotiations, by no means an ardent republican, and, with all this, not a very influential politician after all.

Needless to say, in spite of Mathiez' work,
(15) Danton's statue is still firm on its pedestal, his reputation spotless in the school books of France today. The Third Republic properly made Danton great: in its amoral complacency, he would have been quite at home. Danton always believed that
(20) a popular system of government for his country was absurd; that the people were too ignorant, too inconsistent, and too corrupt to support a legal administration; that, habituated to obey, they required a master. His conduct was in perfect unison
(25) with those beliefs when he acted, but he was too voluptuous for his ambition, too indolent to acquire supreme power. Moreover, his objective seems to have been great wealth rather than great fame.

32. The author's primary purpose in this passage is to

(A) describe and assess the reputation and character of Danton
(B) denounce Danton as one chiefly concerned with enhancing his own notoriety
(C) show the impossibility of drawing an accurate historical picture of a figure like Danton
(D) expose the government leaders' exploitation of Danton's reputation for personal gain
(E) explore the influence of the First French Republic on the Third

33. The tone of the first two sentences of the second paragraph (lines 14-19) can best be described as

(A) honest and straightforward
(B) scholarly
(C) indifferent
(D) ironic
(E) sensational and melodramatic

34. It can be inferred from the passage that Mathiez' findings about Danton were

(A) accepted by the public as accurate
(B) refuted by later writers
(C) largely ignored
(D) attacked as an attempt to embarrass the government
(E) the cause of continuing controversy and debate

35. The author suggests that one of the reasons Danton was a particularly appropriate patron for the Third French Republic was that its people

(A) admired his efforts to structure the government during the First French Republic
(B) shared his distaste for fame and hero worship
(C) bore out his cynical beliefs about the nation
(D) shared his ambition for political power
(E) were driven like him to acquire wealth

GO ON TO THE NEXT PAGE

In connection with the appraisal of mineral resources, the need to differentiate the known and the recoverable from the undiscovered and the uneconomic requires that a resource classification system convey two prime elements of information: one, the degree of certainty about the existence of the materials, and two, the economic feasibility of recovering them. These two elements have generally been recognized in existing terminology, but inconsistently and incompletely. Thus, as used by both the mining and the petroleum industries, the term "reserves" generally refers to economically recoverable material in identified deposits, and the term "resources" includes, in addition, both identified deposits that cannot be profitably recovered now and deposits not yet discovered.

The degree of certainty about the existence of the materials is described by terms such as "proved," "probable," and "possible," the terms traditionally used by industry, and "measured," "indicated," and "inferred," the terms devised during the Second World War by the Geological Survey and the Bureau of Mines to serve the broader purpose of national resource appraisal. Use of these degree-of-certainty terms is by no means standard, but all of their definitions show that they refer only to deposits or structures known or surmised to exist. Moreover, these terms are usually linked to deposits minable at a profit; the classification system comprising these terms thus neglects deposits that might become minable as the result of economic or technological developments.

To remedy these defects, I have suggested that existing terminology be expanded into a broader framework involving both degree of certainty and feasibility of recovery. Either of the series of terms already used to describe degree of certainty may be used with reference to identified deposits and applied not only to presently minable deposits but to others that have been identified with the same degree of certainty. Feasibility-of-recovery categories can be designated by the terms "recoverable," "paramarginal," and "submarginal. . . ." Paramarginal resources are defined here as those that are recoverable if the price of the resource increases to as much as one and one-half times that prevailing now. Over the longer period, we can also expect that technological advances will make it profitable to mine resources that would be even more costly to exploit now. That, of course, is the reason for trying to take account of submarginal resources. It is reasonable to believe that continued technological progress will create recoverable reserves from this category.

36. According to the passage, industry commonly defines "reserves" (line 12) as which of the following?

I. Currently unidentified or economically unrecoverable
II. Existing in known deposits
III. Profitable to develop

(A) I only (B) II only (C) III only
(D) II and III only (E) I, II, and III

37. It can be inferred that the author's "broader framework" (lines 34-35) is most innovative in its treatment of which of the following?

(A) Degree of certainty
(B) Feasibility of recovery
(C) Magnitude of supply
(D) Probability of existence
(E) Margin of error

GO ON TO THE NEXT PAGE

38. The price factor referred to in lines 45-47 reflects the assumption that

(A) technological progress will facilitate differentiation of paramarginal and submarginal resources
(B) increases in the value of a resource may make the mining of its paramarginal deposits economically worthwhile
(C) prices higher than one and one-half times those prevailing will never be acceptable
(D) inflationary price increases of minerals will probably not affect their recoverability
(E) costs of mining submarginal resources vary from place to place

39. Based on the information in the passage, it would be contradictory to use which of the following pairs of terms to label a certain mineral deposit?

(A) Submarginal and proved
(B) Inferred and paramarginal
(C) Indicated and submarginal
(D) Probable and recoverable
(E) Measured and possible

40. The author apparently feels that the terminology traditionally used in connection with resource appraisal is

(A) totally unsuitable
(B) inaccurate and excessively full of jargon
(C) in need of elaboration and standardization
(D) complex to use but appropriate
(E) useful and precise

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 3

Time—30 minutes

25 QUESTIONS

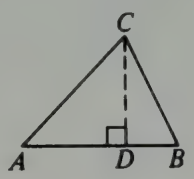
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$

The number of degrees of arc in a circle is 360.

The measure in degrees of a straight angle is 180.



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Definitions of symbols:

$=$	is equal to	$\leqq$	is less than or equal to
$\neq$	is unequal to	$\geqq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

---

1. If $\frac{24 \times 3}{y} = 9$, then $y =$

   (A) $\frac{1}{8}$

   (B) 3

   (C) 8

   (D) 16

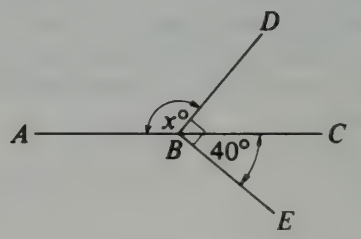
   (E) 24

2. Of the following computations for the volume of a box, which represents the greatest value less than 100 ?

   (A) $2 \times 4 \times 7$
   (B) $2 \times 4 \times 8$
   (C) $3 \times 3 \times 9$
   (D) $3 \times 4 \times 6$
   (E) $3 \times 5 \times 7$

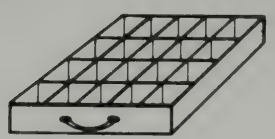
3. The weights of four packages, A, B, C, and D, are in the order $A < B < C < D$. Which of the following could be true about the combined weights of certain of these packages?

   (A) $A + C = B$
   (B) $B + C = A$
   (C) $A + B = C + D$
   (D) $A + D = B + C$
   (E) $A + B + D = C$



4. In the figure above, $DB \perp BE$ and AC is a line segment. What is the value of x ?

   (A) 40 (B) 70 (C) 130 (D) 150 (E) 160



5. The figure above represents one compartmented drawer of a chest of identical drawers. If each compartment in each drawer contains 25 nails and if the chest contains 3,000 nails, exactly how many drawers are in the chest?

   (A) 5 (B) 6 (C) 10 (D) 12 (E) 20

GO ON TO THE NEXT PAGE

6. If x, y, and z are odd numbers, which of the following is an odd number?

(A) $(x + y) + (y + z)$
(B) $x - y$
(C) $x + y + z$
(D) $xy + z$
(E) $(x - y) + (y - z)$

$$\begin{array}{r} \square 4 \\ 3\ \square \\ \square 3 \\ 5\ \square \\ \square 1 \\ \hline 15\ \square \end{array}$$

7. In the addition problem shown above, $\square$ represents the same digit in each number. What must $\square$ represent in order to make the answer correct?

(A) 8 (B) 6 (C) 5 (D) 4 (E) 2

---

8. On a certain map, two points 2.4 inches apart indicate two towns 12 miles apart. How long, in miles, is a straight road which measures 0.2 inch on the map?

(A) 1.0
(B) 2.5
(C) 5.0
(D) 10.0
(E) 12.0

9. If $y = 6x$ and $x = 2z$, what is y in terms of z?

(A) $\frac{1}{3}z$

(B) $3z$

(C) $4z$

(D) $8z$

(E) $12z$

10. Let m equal the greatest possible 3-digit number in which no digit is repeated. Let n equal the least positive 3-digit number that can be made using all of the digits of m. What is the value of $m - n$?

(A) 198
(B) 222
(C) 864
(D) 885
(E) 888

11. If $y = \frac{1}{2}$, then $y^2 - y + \frac{1}{4} =$

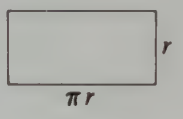
(A) 0 (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) $\frac{3}{4}$ (E) 1

12. If $\frac{3}{4}x = \frac{6}{8}x$, what is the value of x?

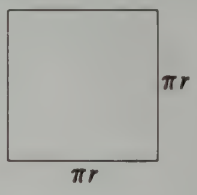
(A) 0 (B) 1 (C) 4 (D) 8
(E) It cannot be determined from the information given.

13. Which of the following rectangles, not necessarily drawn to scale, has an area equal to that of a circle of radius r?

(A)



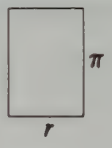
(B)



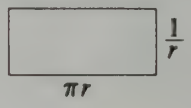
(C)



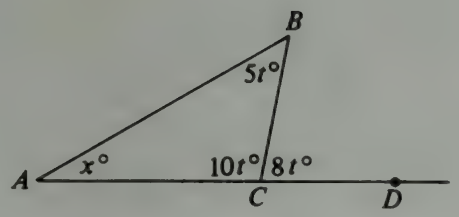
(D)



(E)



GO ON TO THE NEXT PAGE



14. In the figure above, ACD is a line segment. What is the value of x ?

(A) 30 (B) 33 (C) 36 (D) 40 (E) 45

Questions 15-16 refer to the operation represented by ∇ and defined by the equation $x \nabla y = x + y + xy$ for all numbers x and y; for example, $(-6) \nabla (2) = (-6) + (2) + (-12) = -16$.

15. $\left(-\frac{1}{2}\right) \nabla 3 =$

(A) -5

(B) $-\frac{3}{2}$

(C) 1

(D) 4

(E) 5

16. If $8 \nabla k = 3$, then $k =$

(A) -5

(B) $-\frac{5}{9}$

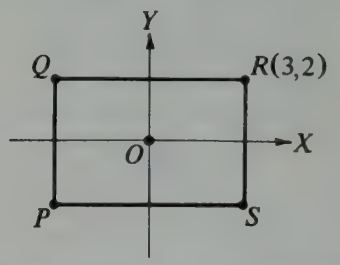
(C) $\frac{3}{8}$

(D) $\frac{5}{9}$

(E) 5

17. If the first question in Part B of a test is number 45, the last question in Part B is number 95, and, in general, the nth question in Part B is number $(44 + n)$, how many questions are there altogether in Part B of the test?

(A) 59
(B) 51
(C) 50
(D) 49
(E) 40



18. In the figure above, the sides of rectangle $PQRS$ are parallel to the X- and Y-axes as shown. If the rectangle is rotated clockwise about the origin until vertex R is on the positive X-axis, what will be the new coordinates of R ?

(A) $(2, 0)$ (B) $(\sqrt{5}, 0)$ (C) $(3, 0)$
(D) $(\sqrt{13}, 0)$ (E) $(5, 0)$

19. After the 1st bounce, a certain ball always bounces $\frac{2}{3}$ of the height of its previous bounce. If the height of its 5th bounce is 1 meter, the height of its 3rd bounce in meters was

(A) $\frac{4}{9}$

(B) $1\frac{1}{2}$

(C) 2

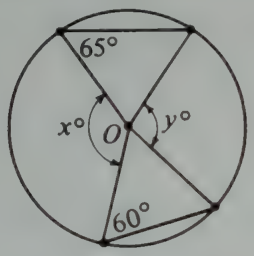
(D) $2\frac{1}{4}$

(E) $2\frac{1}{2}$

GO ON TO THE NEXT PAGE

20. After Jane gave Bill \$4, she then had \$12 more than Bill. How much more money than Bill did Jane have originally?

(A) \$4
(B) \$8
(C) \$12
(D) \$16
(E) \$20



21. In the circle above with center O, $x + y =$

(A) 110 (B) 125 (C) 180
(D) 240 (E) 250

---

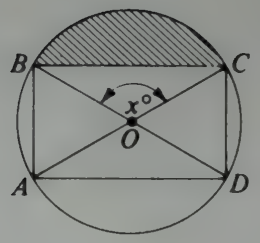
22. On a certain test, a class with 10 students had an average (arithmetic mean) score of 60 and a class with 15 students had an average score of 80. What was the average score on that test for the 25 students?

(A) 70 (B) 71 (C) 72 (D) 75
(E) It cannot be determined from the information given.

23. If $x^3y^2z < 0$, which of the following must be true?

I. $xz < 0$
II. $yz < 0$
III. $xyz < 0$

(A) None
(B) I only
(C) III only
(D) I and II
(E) II and III



24. In the figure above, the vertices of rectangle $ABCD$ lie on a circle with center O. If $BD = 8$, $CD = 4$, and $x = 120$, what is the perimeter of the shaded region?

(A) $\frac{8}{3}\pi + 2\sqrt{3}$

(B) $\frac{8}{3}\pi + 4\sqrt{2}$

(C) $\frac{8}{3}\pi + 4\sqrt{3}$

(D) $8\pi + 2\sqrt{3}$

(E) $8\pi + 4\sqrt{3}$

---

25. In $2n$ years, Lisa will be $(6m + 1)$ times her current age. In terms of m and n, what is Lisa's current age?

(A) $\frac{n}{3m}$

(B) $\frac{2n}{6m + 1}$

(C) $\frac{2n - 1}{6m - 1}$

(D) $6m - 2n + 1$

(E) It cannot be determined from the information given.

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 5

Time—30 minutes

35 QUESTIONS

In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

The following information is for your reference in solving some of the problems.

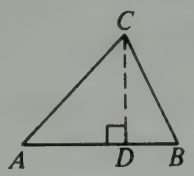
Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$

The number of degrees of arc in a circle is 360.

The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$	is equal to	$\leqq$	is less than or equal to
$\neq$	is unequal to	$\geqq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \dfrac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

---

1. If $0.25 + 0.25 + x = 1$, then $x =$

   (A) 50 (B) 25 (C) $\frac{1}{2}$ (D) $\frac{1}{4}$ (E) $\frac{1}{8}$

2. Which of the following is (are) true?

   I. $12^2 = 144$
   II. $12^3 = 3^{12}$
   III. $12^6 = 72$

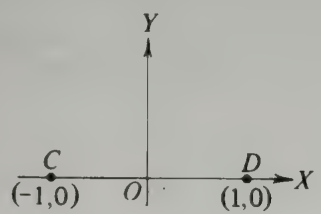
   (A) I only (B) II only (C) III only
   (D) I and III only (E) I, II, and III

3. If 2 out of 30 students in a class are athletes, what fraction of the students in this class are athletes?

   (A) $\frac{1}{60}$ (B) $\frac{1}{28}$ (C) $\frac{1}{16}$ (D) $\frac{1}{15}$ (E) $\frac{1}{14}$

4. Which of the following numbers can be used to show that not all odd numbers are primes?

   (A) 3 (B) 5 (C) 7 (D) 11 (E) 15

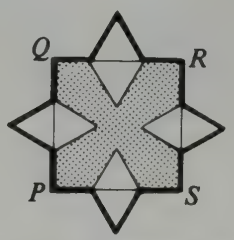


5. Points C and D lie in the XY-plane with coordinates as shown in the figure above. If Q is a point, not shown, equidistant from C and D, then Q can be any point on

   (A) the Y-axis
   (B) the X-axis
   (C) line segment CD
   (D) the line with equation $x = y$
   (E) the line with equation $x = -y$

6. Three persons line up at a ticket window. In how many different orders can they arrange themselves in line?

   (A) 3
   (B) 4
   (C) 6
   (D) 9
   (E) 12



7. In the figure above, all triangles have the same area. If the area of the shaded region is 84 and the area of square $PQRS$ is 100, what is the total area of the regions outlined by the heavy line?

   (A) 16 (B) 92 (C) 108 (D) 116 (E) 132

GO ON TO THE NEXT PAGE

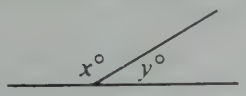
5

Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet blacken space

A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

Notes: 1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. In a given question, a symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as x, n, and k stand for real numbers.

EXAMPLES

	Column A	Column B	Answers
E1.	2×6	$2 + 6$	● Ⓑ Ⓒ Ⓓ
E2.	 $180 - x$	y	Ⓐ Ⓑ ● Ⓓ
E3.	$p - q$	$q - p$	Ⓐ Ⓑ Ⓒ ●

	Column A	Column B
8.	$59{,}000 \div 10$	590×10
9.	$\frac{10x - 60}{2}$	$5x - 30$



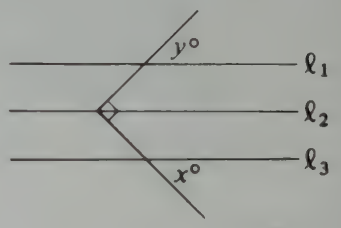
Points X, Y, and Z are on the same line segment.

	Column A	Column B
10.	The distance from X to the midpoint of YZ	60
11.	$(-1)^{87}$	$(-1)^{78}$

$-3 < x < 3$

	Column A	Column B
12.	$-x$	-4

	Column A	Column B



$\ell_1 \parallel \ell_2 \parallel \ell_3$

	Column A	Column B
13.	$x + y$	90
14.	$\left(\frac{1}{2}\right)^4$	$\left(\frac{1}{3}\right)^2$

In a school with 6 classrooms, there are more than 1 and less than 20 students in 1 classroom and exactly 20 students in each of the other 5 classrooms.

	Column A	Column B
15.	Average (arithmetic mean) number of students per classroom	20

$a = -b$

	Column A	Column B
16.	$a + b$	0

$88 < 8x < 100$

	Column A	Column B
17.	x	11.5
18.	$p(r + s) + q(r + s)$	$(p + q)(r + s)$

GO ON TO THE NEXT PAGE

SUMMARY DIRECTIONS FOR COMPARISON QUESTIONS

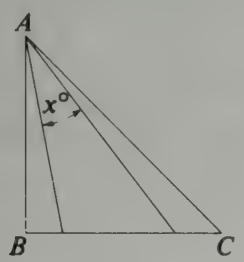
Answer: A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

	Column A	Column B
	$4k = 5m$ $k > 0$	
19.	k	m

A scale drawing is made of the floor of a room.

20.	The perimeter of the drawing when a scale of 1 centimeter to 1 meter is used	The perimeter of the drawing when a scale of 1 centimeter to 0.5 meter is used

	$(x - p)(x + q) = 15$ $x > p$	
21.	x	q



$AB = 3$, $BC = 3$, and $AC = 3\sqrt{2}$

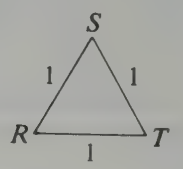
22.	x	45

	Column A	Column B
	In a sequence S of numbers, the first term is 3, the second term is -3, the third term is 3, and so on, with all odd numbered terms equal and all even numbered terms equal. x and y are consecutive terms of S.	
23.	$x + y$	xy

The volume of a cylinder is equal to the area of its base times its height.

24.	The volume of a cylinder with base of radius r and height h	The volume of a cylinder with base of radius $2h$ and height r

	$n > 0$	
25.	16% of n	$\frac{4n}{25}$



26.	The area of $\triangle RST$	0.5

r is an integer greater than 1 and $\boxed{r}$ is defined as the least integer factor of r that is greater than 1.

27.	$\boxed{r}$	$\boxed{r+1}$

GO ON TO THE NEXT PAGE

Solve each of the remaining problems in this section using any available space for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

28. One-eighth of an hour is what fraction of the time between noon on Monday and noon on Wednesday of the same week?

(A) $\frac{1}{16}$ (B) $\frac{1}{192}$ (C) $\frac{1}{384}$

(D) $\frac{1}{392}$ (E) $\frac{1}{476}$

	Event I	Event II	Event III
First Place (5 points)	School A	School B	
Second Place (3 points)		School C	
Third Place (1 point)			

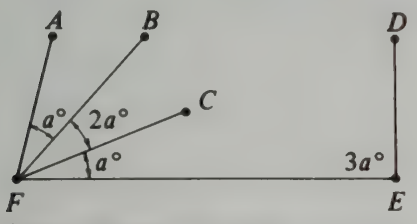
29. Shown above is a partially completed score card for an athletic contest among schools A, B, and C. If each school entered one contestant in each of three events and there were no ties, what is the least possible total score that any one of these schools could achieve for all three events?

(A) 3 (B) 4 (C) 5 (D) 6 (E) 7

30. If x is the average (arithmetic mean) of k and 10, and y is the average of k and 4, what is the average of x and y?

(A) 14 (B) 7 (C) $\frac{k+14}{4}$

(D) $\frac{k+14}{2}$ (E) $\frac{k+7}{2}$



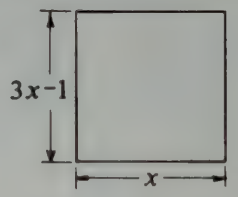
Note: Figure not drawn to scale.

31. If the figure above were redrawn to scale so that $a = 30$, and if segments FA, FB, FC, and ED were extended indefinitely upward, how many points of intersection would there be in addition to E and F?

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

32. Two runners, A and B, pass a post that is 100 meters from the finish line at the same time and then run the final 100 meters of a race at different constant speeds. After the runners pass the post, A runs 80 meters in the same time that B runs 60 meters. How many meters will B have left to run in the race when A reaches the finish line?

(A) 15
(B) 20
(C) 25
(D) 35
(E) 40



33. The figure above is a square. What is its area?

(A) 9

(B) 4

(C) 1

(D) $\frac{1}{4}$

(E) It cannot be determined from the information given.

34. If the cost of p pencils is c cents, what is the cost, in cents, of n pencils at the same rate?

(A) nc (B) $\frac{nc}{p}$ (C) $\frac{pc}{n}$ (D) pn (E) $\frac{pn}{c}$

35. If m is 10 percent greater than y and n is 20 percent greater than x, then mn is what percent greater than xy?

(A) 20%
(B) 30%
(C) 32%
(D) 100%
(E) 200%

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 6

Time—30 minutes

45 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. PRECISE: (A) original (B) inexact (C) obsolete (D) unproductive (E) eventual

2. FATIGUE: (A) pursue (B) revise (C) diet (D) refresh (E) discard

3. INQUISITIVE: (A) undecided (B) incurious (C) defective (D) misinformed (E) distraught

4. FLOURISH: (A) wither (B) relax (C) enact (D) recklessly indulge (E) quietly forego

5. ECSTASY: (A) trickery (B) incompetence (C) uncertainty (D) misconduct (E) wretchedness

6. LISTLESS: (A) energetic (B) unconcerned (C) pathetic (D) hopeful (E) regulated

7. CRASS: (A) free (B) large (C) legal (D) actual (E) delicate

8. REPROOF: (A) guidance (B) praise (C) understanding (D) relief (E) ambition

9. INTREPID: (A) extroverted (B) exclusive (C) fearful (D) baffling (E) adroit

10. DESIST: (A) beg (B) satisfy (C) go on (D) push together (E) look up to

11. PARODY: (A) singularity (B) abstinence (C) monotony (D) reverent imitation (E) unforgivable insult

12. REPLETE: (A) devoid (B) tasteful (C) straight (D) dissimilar (E) untouched

13. BANEFUL: (A) brilliant (B) beneficial (C) mysterious (D) rough (E) careful

14. RHAPSODIZE: (A) paralyze (B) drone (C) perfect (D) ingratiate (E) render

15. RETICENCE: (A) prediction (B) nonsense (C) boredom (D) popularity (E) candor

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ---, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

16. Prolonged secrecy ---- suspicion and allows rumor or false report to harden into accepted fact.

(A) imitates (B) impounds (C) nourishes (D) exorcises (E) invalidates

17. The prime minister presented a --- program that called for radical revision of the constitution without specifying the nature of the ----.

(A) limited. .reaction (B) vague. .changes (C) viable. .sanctions (D) unique. .traditions (E) logical. .connections

18. This interest in the past as a repository of the unfamiliar was parallel to that which prompted the naturalists to collect and exhibit ---- forms of life in the newly established museums.

(A) exotic (B) native (C) modern (D) typical (E) higher

19. Direct and provocative, this volume will be no ---- for those who take books as their opiate, finding passive reading easier than thinking.

(A) goad (B) enigma (C) catalyst (D) sedative (E) challenge

20. Although William the Conqueror preserved whatever institutions of the Anglo-Saxon regime suited his purpose, innovation was more evident than the maintenance of ---- during his reign.

(A) creativity (B) revolt (C) continuity (D) suppression (E) invasion

GO ON TO THE NEXT PAGE

6

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

Despite popular belief, there is nothing especially attractive about a red flag for a bull. Anything waving excites the animal. Apparently, however, a white flag or cape seems brighter to him, and hence is more effective, for the bull is color-blind. Of all the mammals, only humans and some primates enjoy the luxury of color vision. In addition to hundreds of shades of gray, humans can distinguish numerous shades of color. Most mammals see objects only in various shades of gray.

The day-active birds and most reptiles and insects see color, as do a number of fish and crustaceans. But this does not necessarily mean that they see colors as we do. Part of the difference in the colors an animal can see lies in the extent of the spectrum it can perceive. One of the most convincing demonstrations on insects was performed by the British entomologist, H. Eltringham. He caught several of the common butterflies known as tortoise shells, in the pattern of whose wings red is conspicuous. He painted over their compound eyes with a clear red lacquer, fitting them in this way with harmless red glasses. Letting them free, he noted that they flew as well as usual. Apparently the tortoise shell could see with red light, which alone could pass through the red lacquer. When the same test was tried on the white butterfly known as the large cabbage, the insect flew about aimlessly. These observations reflect a logical extrapolation from the normal behavior of the two butterflies, since the tortoise shell commonly visits red flowers, whereas the large cabbage rarely does so. Apparently, the large cabbage sees red flowers as black.

For many insects, the solar spectrum they can perceive includes a very bright band beyond the blue and violet, in the ultraviolet. To a honeybee or housefly, this is the most stimulating part of sunlight. Objects that reflect ultraviolet are far brighter to most insects than objects that do not. Human beings fail to see these differences, since the slightly yellowish lens of our eye absorbs light energy at such short wavelengths.

We can, however, use photographic film sensitive to ultraviolet and learn what an insect sees by means of this component of sunlight. Most things that reflect light visible to us also reflect some ultraviolet. A few provide surprises. The common yellow daisy, for example, absorbs ultraviolet light except at the tips of the petals which are intensely reflecting. In consequence, this flower is visible to insects as a halo of bright ultraviolet spots surrounding the central core where the nectar and pollen wait.

21. Which of the following best expresses the main idea of the passage?

(A) Most animals are color-blind.
(B) Not all animals see color the same way.
(C) Insects can perceive a wider spectrum of colors than can other animals.
(D) Color vision is found only in highly developed animals.
(E) Humans are unable to see what other animals see.

22. It can be inferred from the passage that by daylight a color-blind animal would be able to see which of the following?

(A) Only white objects
(B) Only vague outlines of objects
(C) Only objects that reflect ultraviolet light
(D) Any object, but only if it moved
(E) Any object, but only in shades of gray

23. It can be inferred from the passage that which of the following animals would most likely be able to distinguish colors?

(A) Owl (B) Goat (C) Robin
(D) Mouse (E) Rabbit

24. From which of the following observations could one infer most strongly that an animal can see in red light?

(A) Its body has red markings.
(B) Its only predator is a red-colored bird.
(C) It rarely visits red flowers.
(D) It eats only red-colored grubs.
(E) It lives in a house made of red mud.

25. According to the passage, ultraviolet is very stimulating to most insects because it

(A) appears very bright to them
(B) is reflected only by flowers
(C) has the power to block out other colors
(D) creates a halo of light around an object
(E) is of such rare occurrence

GO ON TO THE NEXT PAGE

Does the totality of a language tell anything about the people who speak it? Consider the English verb "grab." An English speaker says, "I grab it," "I grabbed it," "I will grab it," and so on. Only context tells the listener what it is that is being grabbed and how it is being done. "I grabbed it" is a vague sentence—except in one way. English is remarkably concerned about tense. It insists on knowing whether I grab it now, or grabbed it sometime in the past, or will grab it at a future time. The English language is preoccupied with time; so is the culture of its speakers, who take considerable interest in calendars, record-keeping, history, stock-market forecasts, and the precise time throughout the waking day.

No such statement as "I grabbed it" would be possible in Navajo. In Navajo, tense is of little importance, but the language is considerably more discriminating in other ways. The Navajo language would describe much more about the pronoun "I" of this sentence. The Navajo would not be content merely with "grabbed" either; the verb would have to state more about why and how the act is being done, and whether the thing being grabbed is big or little, animate or inanimate. Further, a Navajo could not simply say "it": the thing being grabbed would have to be described much more precisely and put in a category.

Linguistic studies of the Navajo indicate that the elements that make up their universe are very precisely described. The Navajo language seems to require a perception of the universe that is more exacting than that of English speakers. But is this a true picture of the Navajo? Do they perceive the world any differently from speakers of English? Anthropological and psychological studies of the Navajo show that they visualize themselves as living in an eternal and unchanging universe made up of physical, social, and supernatural forces, among which they try to maintain a balance. To the Navajo, the good life consists of maintaining intact all the complex relationships of the universe. It is obvious that to do so demands a language that makes the most exacting discriminations.

This is not to say that the Navajo hold such a world view because of the structure of their language. It merely suggests that there is an interaction between the language and their culture. Language is more than a way of communicating. It is a living system that is a part of the cultural equipment of a group, and it reveals a culture at least as much as do pottery, kinship groups, and political institutions.

26. The passage as a whole suggests that language provides information about its users by revealing

(A) chiefly their world view
(B) chiefly their concept of time
(C) only the culture of English-speaking peoples
(D) only their environment
(E) only their past experience

27. The author refers to "stock-market forecasts" (lines 13-14) as an example of

(A) a time-oriented concern
(B) a questionable faith in the future
(C) the ambiguities in language
(D) concepts common to all languages
(E) English speakers' avoidance of abstract ideas

28. The author bases the answer to the question "But is this a true picture of the Navajo?" (lines 33-34) primarily on

(A) linguists' analyses of the Navajo language
(B) personal experiences in present-day Navajo societies
(C) comparisons between the English language and the Navajo language
(D) the research of anthropologists and psychologists
(E) depictions of the Navajo in the literature of various cultures

29. The passage implies that the relative unimportance of tense in the Navajo language mirrors the Navajo's

(A) emphasis on abstract ideas
(B) desire for exact description
(C) vision of the universe as eternal and unchanging
(D) complex relationships among kinship groups
(E) preoccupation with keeping records

30. According to the passage, the Navajo regard the complex relationships among the various forces in the universe with

(A) insatiable curiosity
(B) bewildered despair
(C) respectful appreciation
(D) deliberate indifference
(E) aloof skepticism

GO ON TO THE NEXT PAGE

6

Select the word or set of words that best completes each of the following sentences.

31. To the extent that economic change opens new horizons and creates new aspirations and desires without immediate possibilities of fulfillment, it ---- the ---- within a social structure.

(A) alleviates. .problems
(B) increases. .tensions
(C) simplifies. .bureaucracy
(D) reduces. .pressures
(E) emphasizes. .improvement

32. Matisse was unusually ---- about his painting; to read his statements is to see more clearly through his eyes his intents and his technical procedures.

(A) modest (B) evasive (C) superficial
(D) articulate (E) intemperate

33. Some collect rare objects for ---- reasons whereas others ---- their own taste in the pursuit of profitable investment.

(A) aesthetic. .ignore
(B) sentimental. .misconstrue
(C) various. .expand
(D) personal. .trust
(E) valid. .refine

34. It has been ---- that the intelligence of various species of animals differs only in degree, but new experiments show that the differences are ----.

(A) argued. .irrefutable
(B) advanced. .objective
(C) observed. .recognizable
(D) doubted. .questionable
(E) assumed. .qualitative

35. The nostalgic associations we bring to this revived musical can give it an ----, though still affecting, poignancy; however, they cannot quite cover up its flaws.

(A) unsettling (B) incisive (C) unflinching
(D) internal (E) unearned

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

36. PLUG : SOCKET :: (A) key : lock
(B) chair : desk (C) wire : electricity
(D) current : switch (E) fan : wind

37. RAGE : ANGER :: (A) bliss : apathy
(B) fear : shame (C) delight : pleasure
(D) frustration : patience (E) approval : censure

38. VERTEX : PYRAMID :: (A) strand : hair
(B) rectangle : box (C) rung : ladder
(D) frame : picture (E) summit : mountain

39. MOVIE : DIRECTOR :: (A) store : salesperson
(B) business : manager (C) book : critic
(D) competition : athlete (E) product : advertiser

40. CUTLERY : KNIFE :: (A) machinery : fuel
(B) lumber : saw (C) suitcase : handle
(D) bookcase : volume (E) furniture : chair

41. ASPIRANT : AMBITION :: (A) mentor : malice
(B) renegade : faith (C) virtuoso : innocence
(D) rebel : defiance (E) antagonist : maturity

42. ANARCHISM : GOVERNMENT ::
(A) paternalism : ancestors (B) pacifism : war
(C) realism : progress (D) capitalism : commerce
(E) socialism : society

43. STUPIDITY : DUNCE :: (A) obstinacy : introvert
(B) flattery : sycophant (C) fear : clown
(D) acuity : hypocrite (E) deceit : vagrant

44. REMISSION : DISEASE ::
(A) recollection : imagination
(B) resurgence : determination
(C) prescription : diagnosis
(D) quiescence : storm
(E) regression : violence

45. DISCUSSION : ALTERCATION ::
(A) planning : action
(B) speech : tirade
(C) uncertainty : change
(D) dialogue : conversation
(E) group : convention

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

Correct Answers for Scholastic Aptitude Test
Form Code 2F

VERBAL		MATHEMATICAL	
Section 1	Section 6	Section 3	Section 5
1. C	1. B	1. C	1. C
2. A	2. D	2. C	2. A
3. A	3. B	3. D	3. D
4. C	4. A	4. C	4. E
5. E	5. E	5. B	5. A
6. C	6. A	6. C	6. C
7. B	7. E	7. E	7. D
8. A	8. B	8. A	*8. C
9. C	9. C	9. E	*9. C
10. A	10. C	10. A	*10. A
11. D	11. D	11. A	*11. B
12. B	12. A	12. E	*12. A
13. E	13. B	13. A	*13. C
14. B	14. B	14. A	*14. B
15. E	15. E	15. C	*15. B
16. C	16. C	16. B	*16. C
17. B	17. B	17. B	*17. D
18. D	18. A	18. D	*18. C
19. A	19. D	19. D	*19. A
20. D	20. C	20. E	*20. B
21. D	21. B	21. E	*21. D
22. B	22. E	22. C	*22. B
23. E	23. C	23. B	*23. A
24. D	24. D	24. C	*24. D
25. D	25. A	25. A	*25. C
26. C	26. A		*26. B
27. B	27. A		*27. D
28. D	28. D		28. C
29. A	29. C		29. C
30. C	30. C		30. E
31. E	31. B		31. B
32. A	32. D		32. C
33. D	33. A		33. D
34. C	34. E		34. B
35. C	35. E		35. C
36. D	36. A		
37. B	37. C		
38. B	38. E		
39. E	39. B		
40. C	40. E		
	41. D		
	42. B		
	43. B		
	44. D		
	45. B		

*Indicates four-choice questions. (All of the other questions are five-choice.)

The Scoring Process

Machine-scoring is done in three steps:

- *Scanning.* Your answer sheet is "read" by a scanning machine and the oval you filled in for each question is recorded on a computer tape.
- *Scoring.* The computer compares the oval filled in for each question with the correct response. Each correct answer receives one point; omitted questions do not count toward your score. For each wrong answer, a fraction of a point is subtracted to correct for random guessing. For questions with five answer choices, one-fourth of a point is subtracted for each wrong response; for questions with four answer choices, one-third of a point is subtracted for each wrong response. The SAT-verbal test has 85 questions with five answer choices each. If, for example, a student has 44 right, 32 wrong, and 9 omitted, the resulting raw score is determined as follows:

  $$44 \text{ right} - \frac{32 \text{ wrong}}{4} = 44 - 8 = 36 \text{ raw score points}$$

  Obtaining raw scores frequently involves the rounding of fractional numbers to the nearest whole number. For example, a raw score of 36.25 is rounded to 36, the nearest whole number. A raw score of 36.50 is rounded upward to 37.
- *Converting to reported scaled score.* Raw test scores are then placed on the College Board scale of 200 to 800 through a process that adjusts scores to account for minor differences in difficulty among different editions of the test. This process, known as equating, is performed so that a student's reported score is not affected by the edition of the test taken nor by the abilities of the group with whom the student takes the test. As a result of placing SAT scores on the College Board scale, scores earned by students at different times can be compared. For example, an SAT-verbal score of 400 on a test taken at one administration indicates the same level of developed verbal ability as a 400 score obtained on a different edition of the test taken at another time.

How to Score the Test

You can verify the College Board SAT scores reported to you recently by using the information in this booklet along with the copy of your answer sheet. *Before you begin, check that the first two characters (number and letter) of the form code you marked in item 3 on your answer sheet are the same as the form code printed on the front of this booklet.* Compare the responses shown on the copy of your answer sheet with the list of correct answers.

SAT-Verbal Sections 1 and 6

Step A: Count the number of correct answers for *section 1* and record the number in the space provided on the worksheet on the next page. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for *section 6* and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: To obtain C, add subtotal A to subtotal B, keeping any decimals. Enter the resulting figure on the worksheet.

Step D: To obtain D, your raw verbal score, round C to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step E: To find your reported SAT-verbal score, look up the total raw verbal score you obtained in step D in the conversion table on page 200. Enter this figure on the worksheet.

SAT-Mathematical Sections 3 and 5

Step A: Count the number of correct answers and the number of incorrect answers for *section 3* and record the numbers in the spaces provided on the worksheet. To determine the subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for the *five-choice questions (questions 1 through 7 and 28 through 35) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: Count the number of correct answers and the number of incorrect answers for the *four-choice questions (questions 8 through 27) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal C, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{3} = \text{subtotal C}$$

Step D: To obtain D, add subtotal A, subtotal B, and subtotal C, keeping any decimals. Enter the resulting figure on the worksheet.

Step E: To obtain E, your raw mathematical score, round D to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step F: To find your reported SAT-mathematical score, look up the total raw mathematical score you obtained in E in the conversion table on page 200. Enter this figure on the worksheet.

SAT SCORING WORKSHEET

FORM CODE 2F

SAT-Verbal Sections

A. Section 1: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal A

B. Section 6: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal B

C. Total unrounded raw score (Total A + B) _____ C

D. Total rounded raw score (Rounded to nearest whole number) _____ D

E. SAT-verbal reported scaled score (See the conversion table on the back cover.) [ ] SAT-verbal score

SAT-Mathematical Sections

A. Section 3: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal A

B. Section 5: Questions 1 through 7 and 28 through 35 (5-choice) _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal B

C. Section 5: Questions 8 through 27 (4-choice) _____ (no. correct) − ⅓ (_____) (no. incorrect) = _____ subtotal C

D. Total unrounded raw score (Total A + B + C) _____ D

E. Total rounded raw score (Rounded to nearest whole number) _____ E

F. SAT-mathematical reported scaled score (See the conversion table on the back cover.) [ ] SAT-math score

Score Conversion Table
Scholastic Aptitude Test
Form Code 2F

Raw Score	College Board Reported Score		Raw Score	College Board Reported Score	
	SAT-Verbal	SAT-Math		SAT-Verbal	SAT-Math
85	800		40	460	620
84	790		39	450	610
83	780		38	440	600
82	770		37	440	590
81	760		36	430	580
80	750		35	430	570
79	740		34	420	570
78	730		33	410	560
77	720		32	410	550
76	710		31	400	540
75	700		30	390	530
74	690		29	390	520
73	680		28	380	520
72	670		27	380	510
71	670		26	370	500
70	660		25	360	490
69	650		24	360	480
68	650		23	350	470
67	640		22	340	460
66	630		21	340	460
65	620		20	330	450
64	610		19	320	440
63	610		18	320	430
62	600		17	310	420
61	590		16	300	410
60	590	800	15	300	400
59	580	790	14	290	390
58	570	780	13	280	390
57	560	770	12	270	380
56	560	760	11	270	370
55	550	750	10	260	360
54	550	740	9	250	350
53	540	730	8	250	340
52	530	720	7	240	330
51	530	710	6	230	320
50	520	700	5	220	310
49	510	700	4	220	300
48	510	690	3	210	290
47	500	680	2	200	280
46	490	670	1	200	270
45	490	660	0	200	260
44	480	650	– 1	200	250
43	480	640	– 2	200	240
42	470	630	– 3	200	230
41	460	630	– 4	200	220
			– 5	200	210
			– 6 or below	200	200